Active inference

# Active inference as a computational framework for modeling empirical data

## FREE ENERGY PRINCIPLE

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# Markov decision processes

For modeling perception, learning, and decision making

- MATLAB (provided in the **appendices** and **supplementary code**) to build active inference (POMDP) models
- https://github.com/rssmith33/Active-Inference-Tutorial-Scripts

- python implementation of active inference that can be found at: https://github.com/infer-actively/pymdp).

- They focus is specifically on the POMDP formulation, which models **time in discrete steps** and **treats beliefs and actions as discrete categories** (referred to as 'discrete state–space' models with 'discrete time')

Active inference

# What is active inference?

- Theoretical framework
- It's a model of how agents (which could be biological organisms or artificial systems) might perceive and act within their environments.
- Agent seeks to minimize surprise or prediction errors about its sensory inputs

**Inference**

**Action**

**Learning**

**Policies**

Decision-making, Perception, and Action selection

# What is active inference?

Active inference, initially a concept within the realm of predictive motor control and later expanded into predictive decision-making, is fundamentally rooted in Bayesian inference and probability theory.

- **Predicting Future Observations**

If I go inside, then I predict I will feel warm

- **Combining Predictions with Preferences**

In want to feel warm, therefore I will go inside

- **Information-Seeking Actions**

I am going to turn on my flashlight because I predict this will help me figure out how to get inside.

Active inference

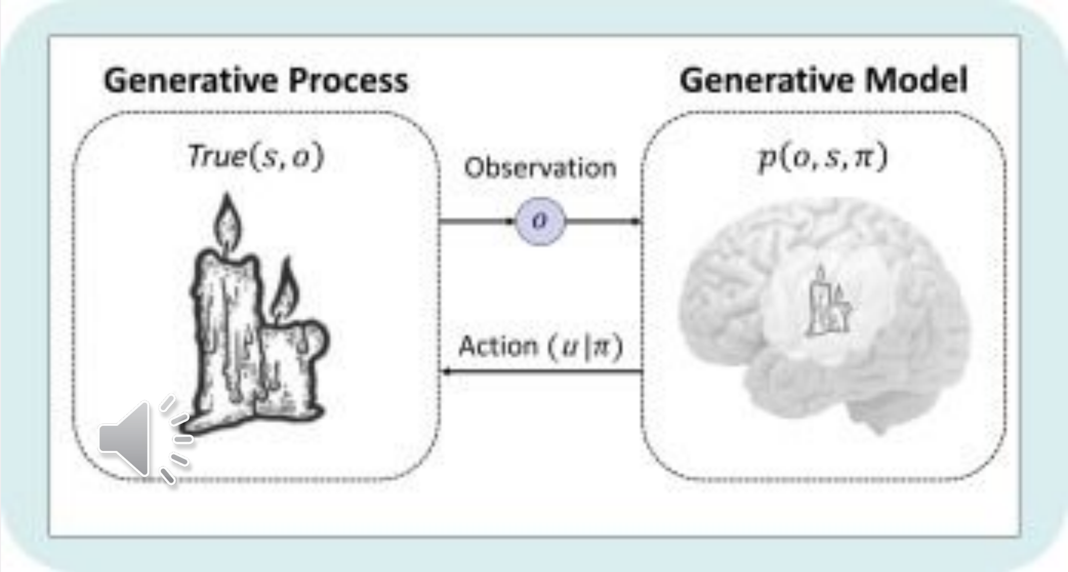
# Generative Model vs. Generative Process

Observation (o)

Hidden states (s) (its hidden because inferred from observations)

Policies (π) which are sequences of actions

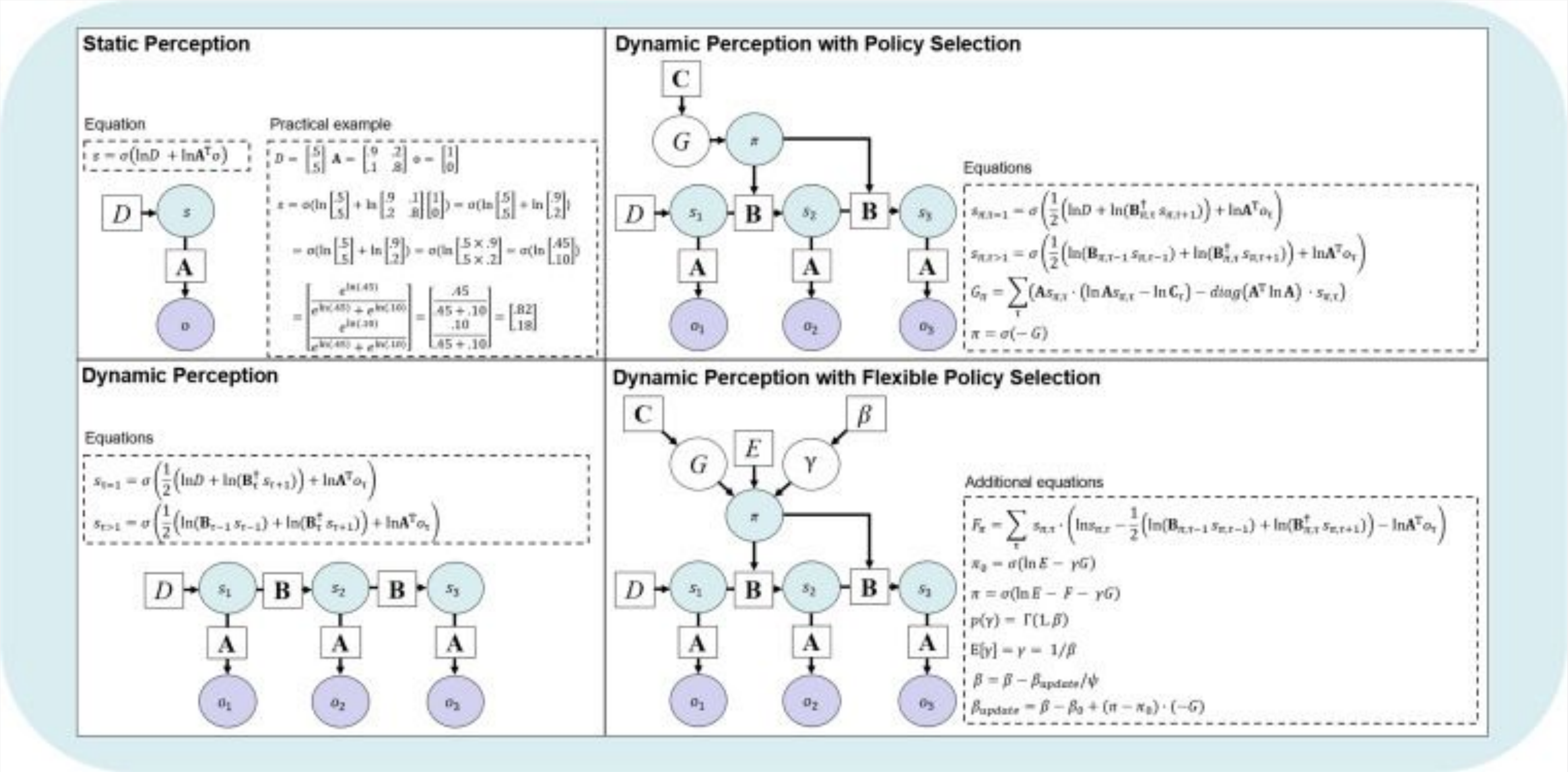
Actions (u)




**Generative Model:** It's a probabilistic representation of an agent's beliefs about the world. This model encapsulates the joint probability of various combinations of observations, hidden states, and policies. **Generative Process:** This refers to the actual objects, events, or processes in the external environment that generate the observations. These are the true, underlying factors that cause the observed sensory data.

| variational free energy (VFE) | expected free energy (EFE) |
|---|---|
| The active inference framework is based on the premise that perception and learning can be understood as minimizing a quantity known as variational free energy (VFE), | and that action selection, planning, and decision-making can be understood as minimizing expected free energy (EFE), which quantifies the VFE of various actions based on expected future outcomes. |

Active inference




Active inference

# Active inference formalism

Based on partially observable Markov Decision Processes (POMDPs)

- The ideal here is that its modeling hidden state of the world that evolve over time

The agent doesn't have full knowledge of the environment

- It must infer the 'hidden' states of the world based on incomplete information contained in observations

Infer the polices (action sequences) most likely to generate preferred observations

**Policies in Reinforcement Learning:**

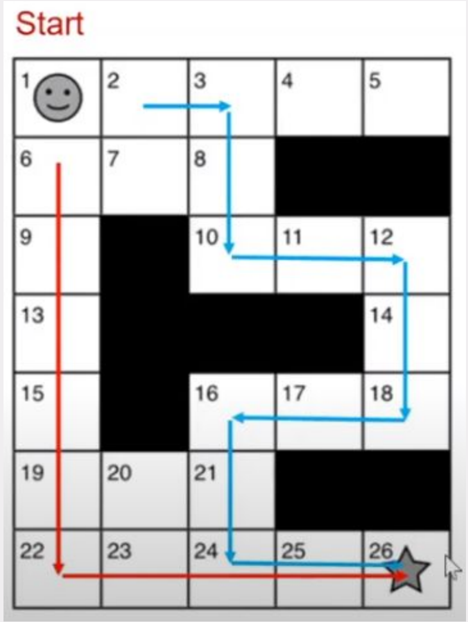
**Definition:** In reinforcement learning, a 'policy' typically denotes a mapping from states to actions, defining what action to take when in a particular state.

**Example:** Suppose an AI is navigating a maze. If it's in state #1 (near a wall), the policy dictates it should move left; if in state #2 (open space), the policy instructs it to move right.

**Policies in Active Inference:**

**Definition:** In active inference, a 'policy' represents a sequence of actions predetermined to a specific depth or number of actions.

**Example:** Consider an AI agent trying to reach a goal. Policy #1 could be a sequence of actions like left, left, right, evaluating the consequence of these actions; Policy #2 might be right, right, left, another sequence the agent could follow.




- O= observation (also called outcomes)
- S= hidden states (hidden because they must be inferred from observations)
- π = policies (possible sequences of actions) u=actions

We evaluate every things based on polices, because the goal is to choose a policy

- P(S| π)= prior belief
- P (O|S, π)= likelihood mapping
- P(S|O, π)= posterior belief

Bayes theorem

$$p(s|o,m)=\frac{p(o|s,m)p(s|m)}{p(o|m)}$$

what we want to figure out in perception

- $P(o|\pi)$= predictive posterior” over observations

The observationa I expect if I choose this policy

- $P(o|c)$= prior preference’ distribution

The observations I find rewarding (higher probability= more rewarding)

Unique element in active inference: uses something with the form a probability distribution to encode reward

use as a matrix essentially that encodes what observations we want over others

- q= denotes approximate distributions

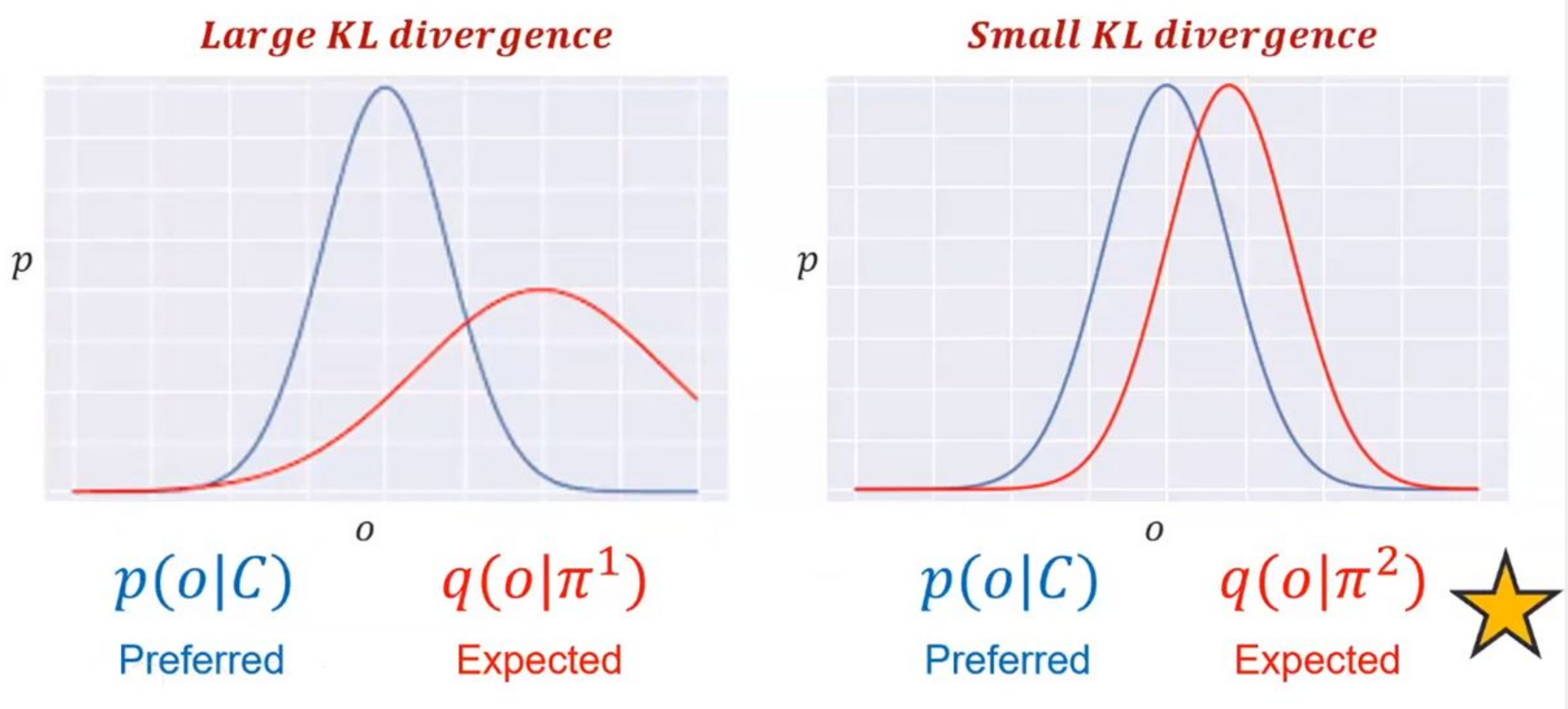
Because exact Bayesian inference is often intractable

- $q(s|\pi)$=approximate posterior over states (best guess)

We want to get this to match $p(s|o, \pi)$ as closely as possible

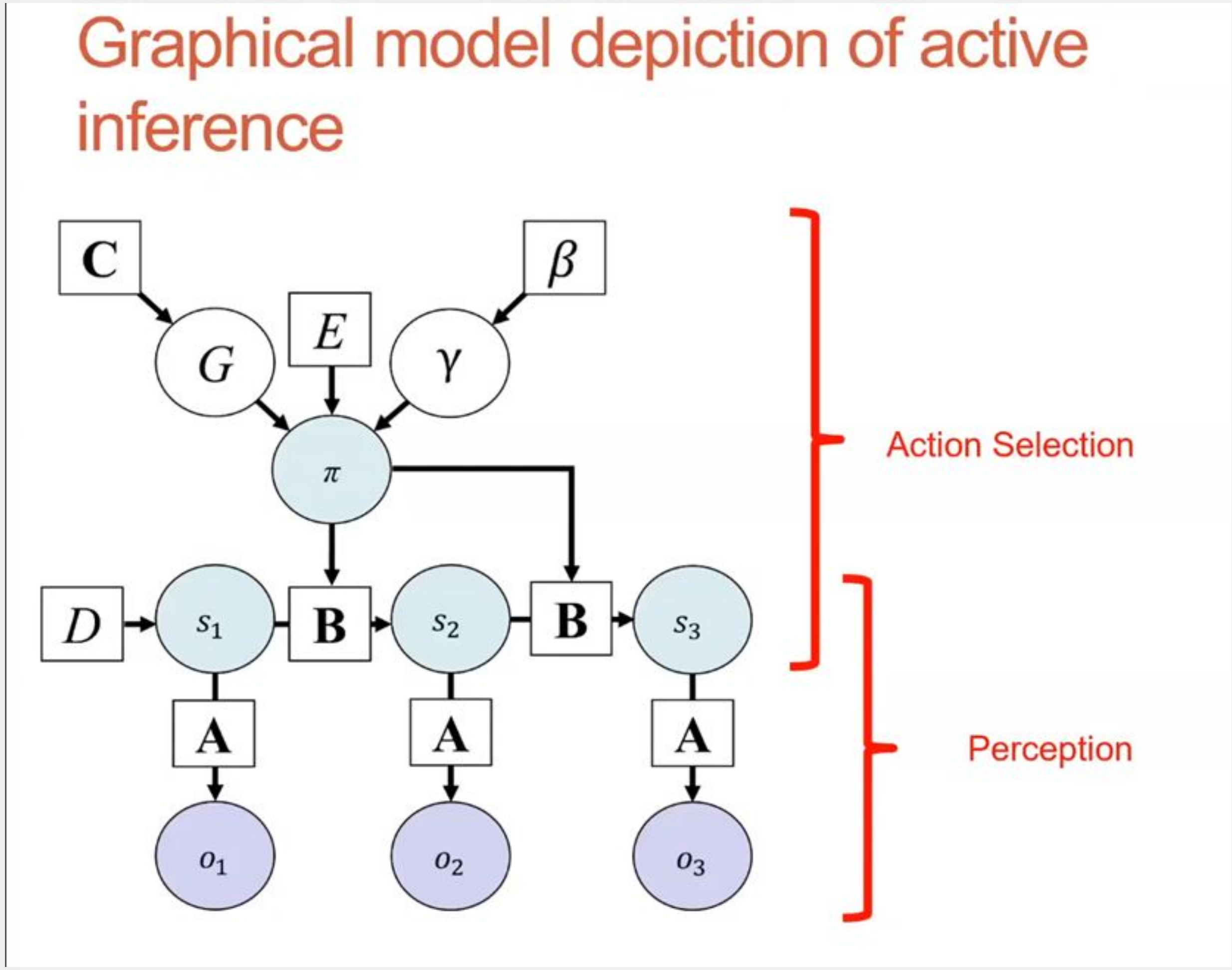
- $q(o|\pi)$= approximate prediction about what I will observe if I choose this policy

- $D_{KL}[p(x)||p(y)]$ – the *Kullback–Leibler (KL) divergence*
  - Smaller values indicate that two **distributions are more similar**
  - **Action selection**: the closer we can get $q(o|\pi)$ to match $p(o|C)$, the smaller their $D_{KL}$ will be (i.e., *we're more likely to observe what we want*)



| Parameter | *D* (Initial State Prior) | C (Prior Preferences over Observations) |
|---|---|---|
| Definition | Represents initial beliefs about hidden states | Represents preferences over observations |
| Use | Influences initial assumptions about the agent's location or state | Influences the agent's preference for certain observations |
| Example | *D*=[0.25,0.25,0.25,0.25] (Uniform belief about starting positions in a maze) | C=[0.7,0.3] (Prefers brighter environments over darker ones) |
| Significance | Affects where the agent thinks it might be initially | Affects what observations the agent prefers or finds more rewarding |

# Bayesian networks




- Here arrows denote conditional dependencies
- Circle correspond to random variables that are updated during inference

  S=q(s), π=q(π)
- Squares indicate fixed parameters

- Updated more slowly through learning

Bottom part: the states and observations  corresponds to perception in active inference
Top part: policies and the things that policies depend on is the action selection part

## Static and dynamic perception in discrete
## State spaces and discrete time

**Task instructions**: "On each trial you have the choice of two slot machines. One will pay out $4, and the other will pay out $0. You can either guess right away or you can ask for a hint about which one is more likely to pay out. However, if you ask for the hint, you can only win $2."



**Example states and outcomes**

**Hidden State Factor 1: Context**
- Left machine is more likely to pay out
- Right machine is more likely to pay out

**Hidden State Factor 2: Choice states**
- Start
- Ask for the hint
- Choose left machine
- Choose right machine

**Outcome Modality 1: Hint**
- No hint
- Left machine is more likely
- Right machine is more likely

**Outcome Modality 2: Outcome**
- Start
- Lose
- Win

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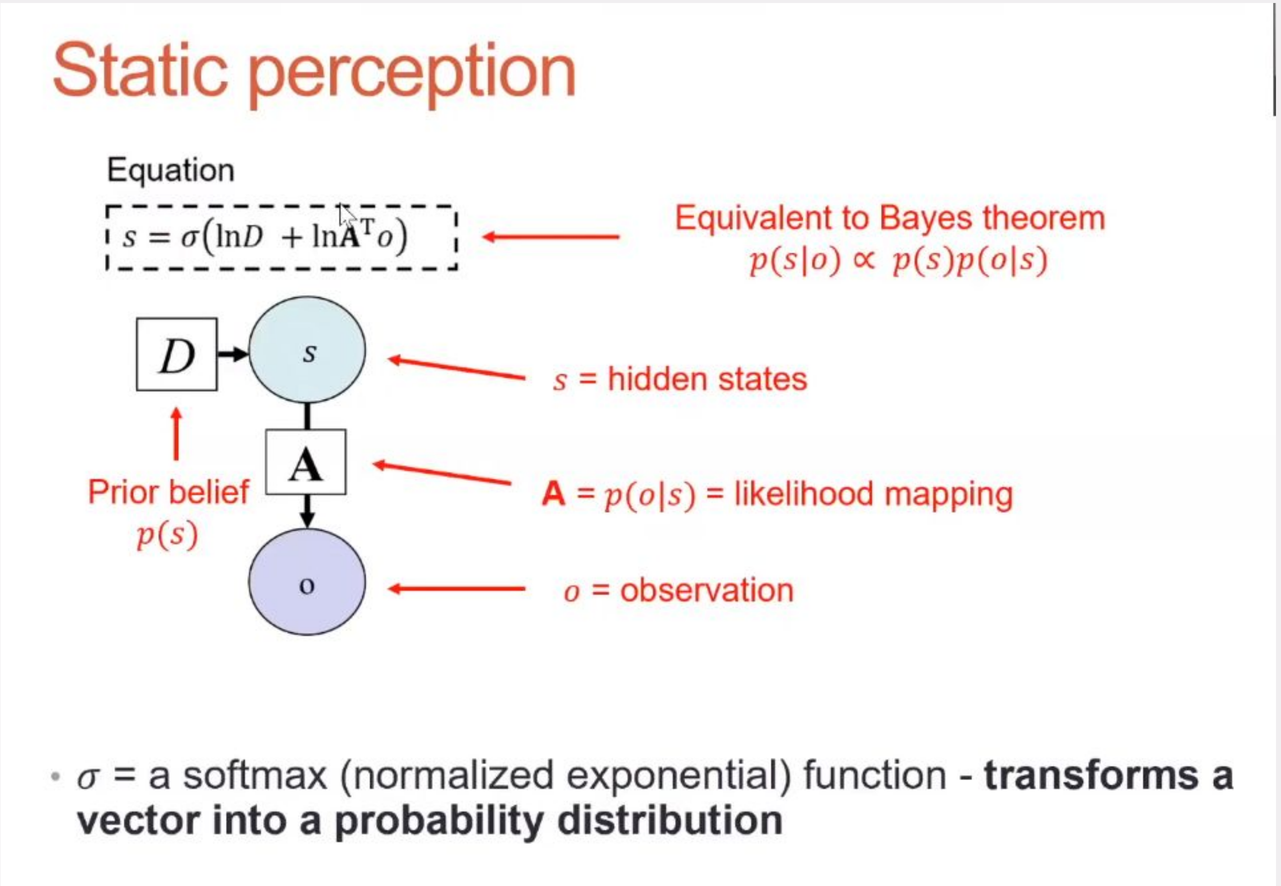
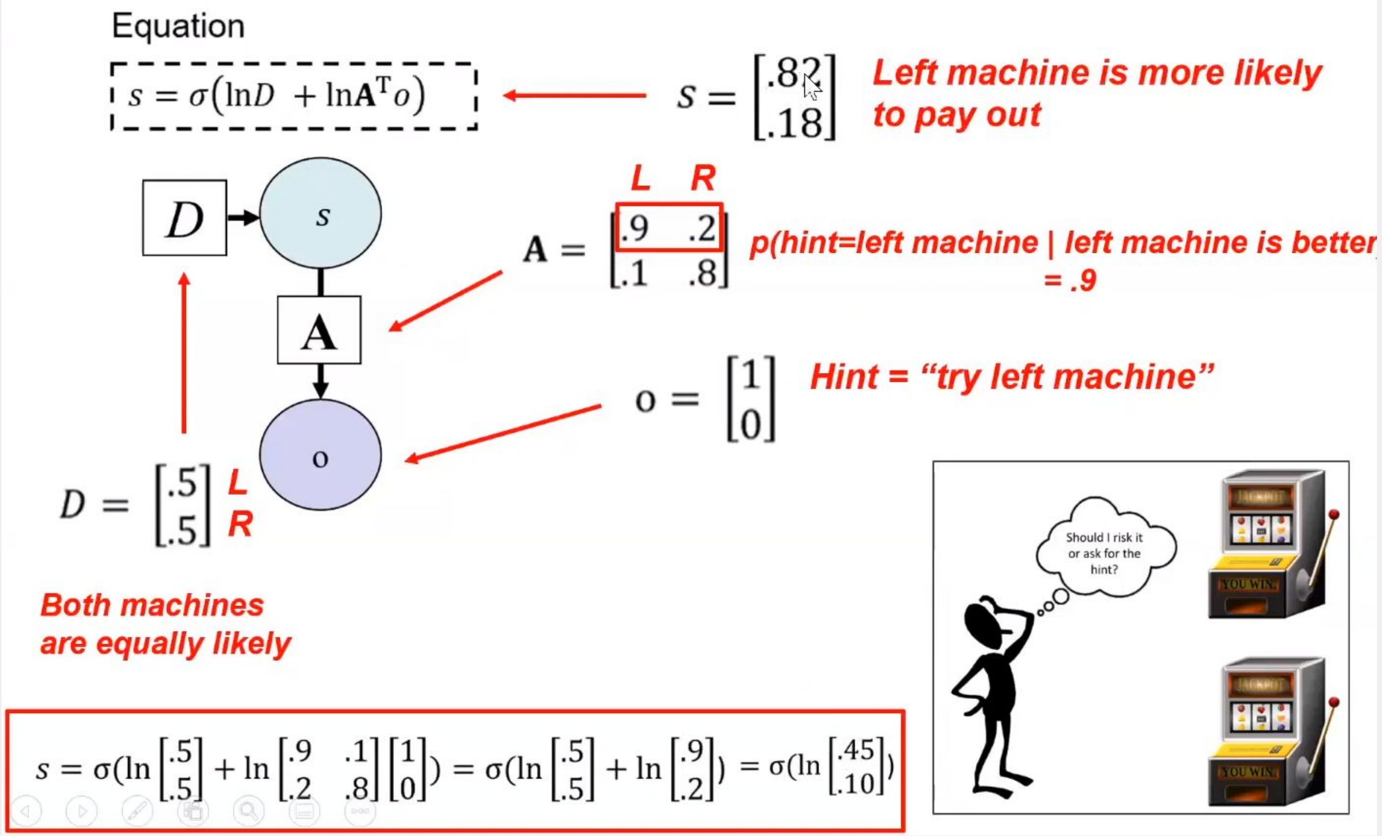
# Static perception

Graphic model we have just observations



# Static perception



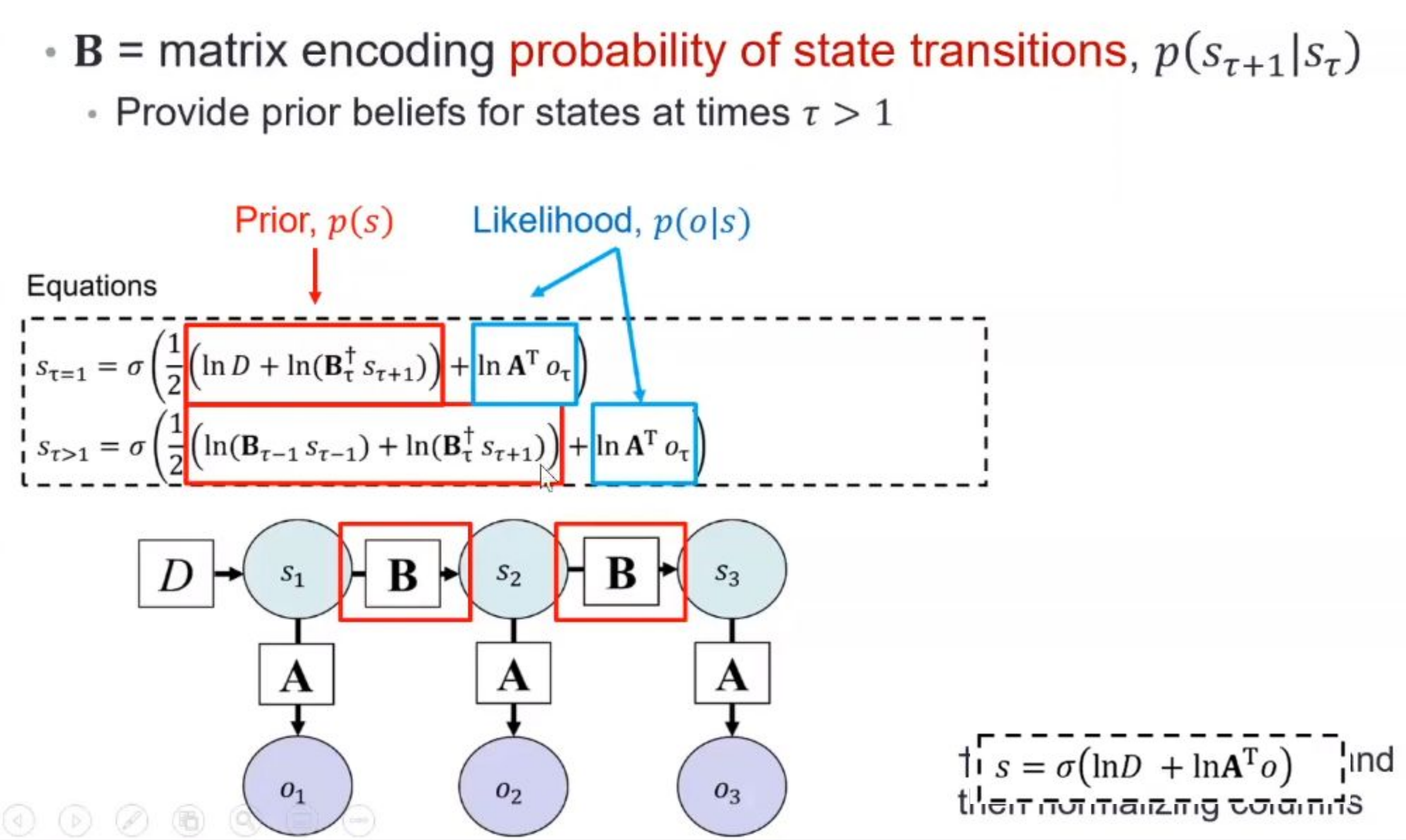
# Dynamic perception: Hidden Markov model

There's different types of time in active inference

- $t$: Denotes the time points at which each new observation is presented.
- $T$: Denotes time points within each trial, sometimes referred to as 'epochs,' about which
- **Retrospective Inference:** The ability to update beliefs about past states ($T$) based on new observations in the present ($t$).
- **Prospective Inference:** The ability to update beliefs about future states ($T$) based on new observations in the present ($t$).
- **Example:** In an 'explore–exploit' task, observing a hint at one time point ($t$) could update beliefs about which slot machine will be better at the next time point ($T$).

# Dynamic perception: Hidden Markov model

We are connecting the states over time with these B (matrices here that just encode the probability of moving some state given that I was some state at a previous time )




Static and dynamic perception

Active inference formalism

Active inference

# Dynamic perception: Hidden Markov model

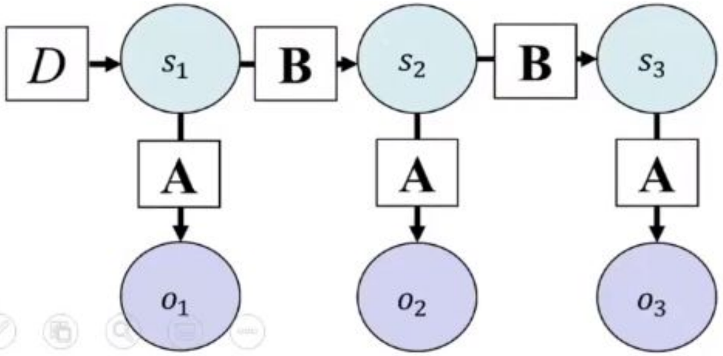
- Allows you to update beliefs **about earlier time points** when getting observations **at later time points** ($\tau$ vs. $t$)

Prior from the future

Equations

$$s_{\tau=1} = \sigma\left(\frac{1}{2}\left(\ln D + \boxed{\ln(\mathbf{B}_\tau^\dagger\, s_{\tau+1})}\right) + \ln \mathbf{A}^\mathrm{T}\, o_\tau\right)$$

$$s_{\tau>1} = \sigma\left(\frac{1}{2}\left(\ln(\mathbf{B}_{\tau-1}\, s_{\tau-1}) + \boxed{\ln(\mathbf{B}_\tau^\dagger\, s_{\tau+1})}\right) + \ln \mathbf{A}^\mathrm{T}\, o_\tau\right)$$

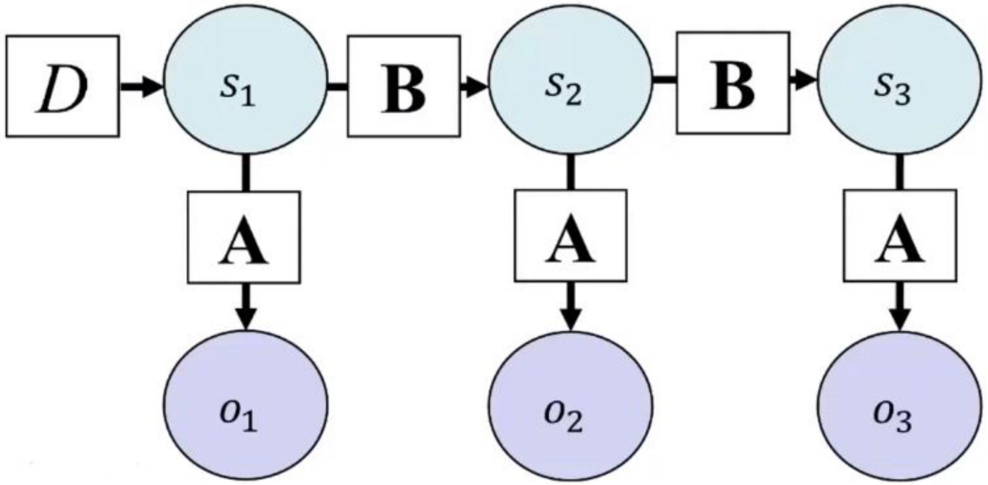



† indicates a transpose and then normalizing columns

# Variational (marginal) message passing

In this case with active inference the main way that we tend to do it is with varitional message passing or updated version called marginal message passing but that’s what the equations showed us implement

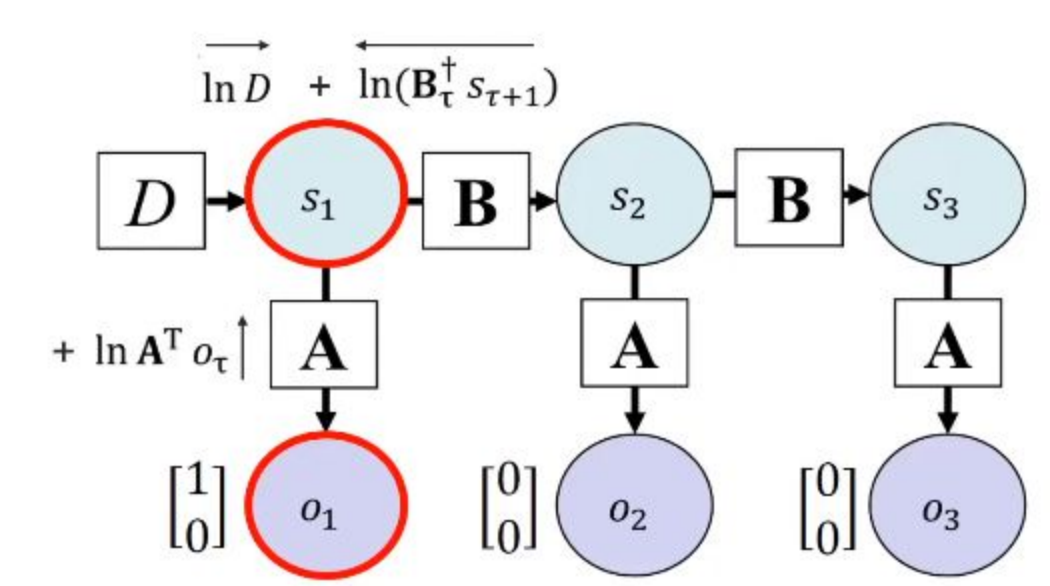
- These equations implement **variational message passing**
  - An efficient way to do *approximate* **posterior inference on a graph**
  - Based on **minimizing** a quantity called **variational free energy ($F$)**
    - Will **first illustrate how this works** and **then introduce $F$ formally**



# How this shape work and what is F?

Encoding observation is one so fixing whatever observation is a tau equals one and so lets say its observation the top observation □ o1 and now what you would do is you would use that and first try update your belief about whatever states are for tau=1, we do that by passing these messages so in other words in this case its just the log (the natural log of the prior belief) (the natural log of the prior belief from the future time point) and then our likelihood

**Variational message passing**



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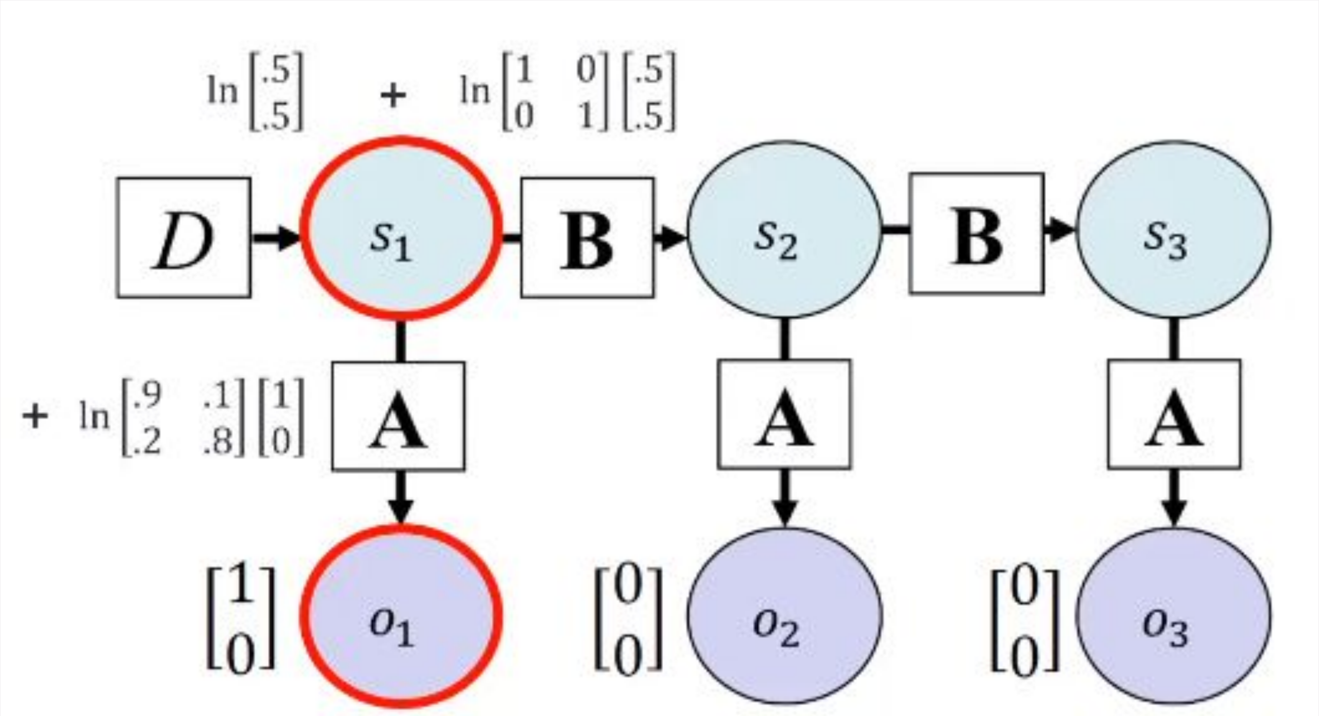
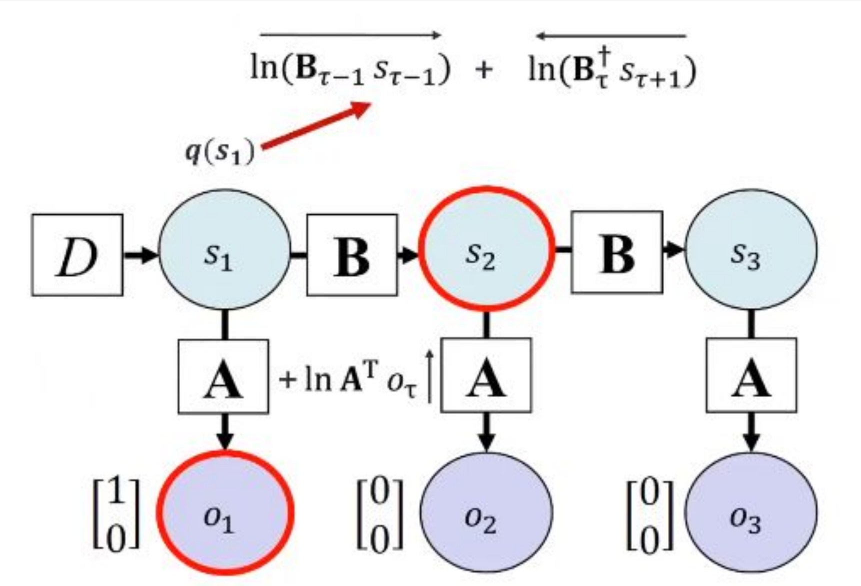
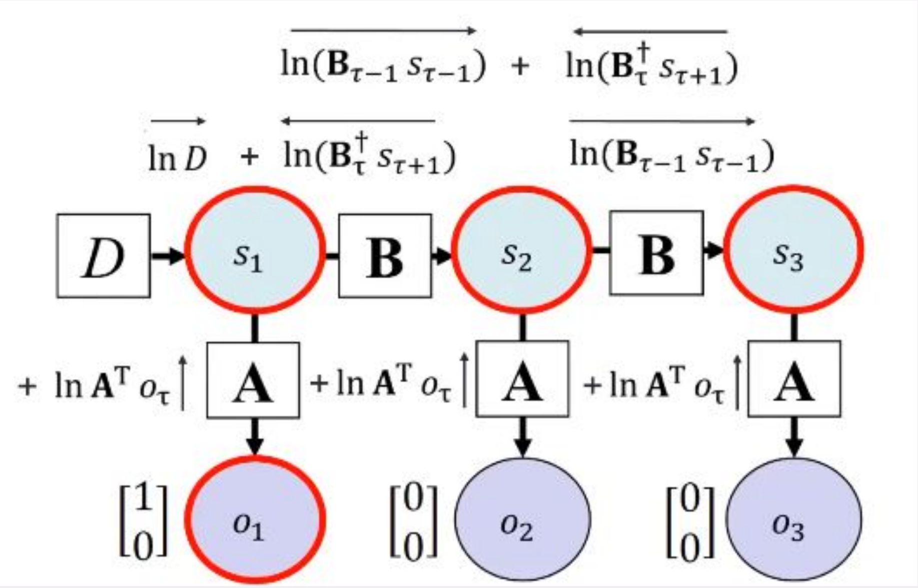
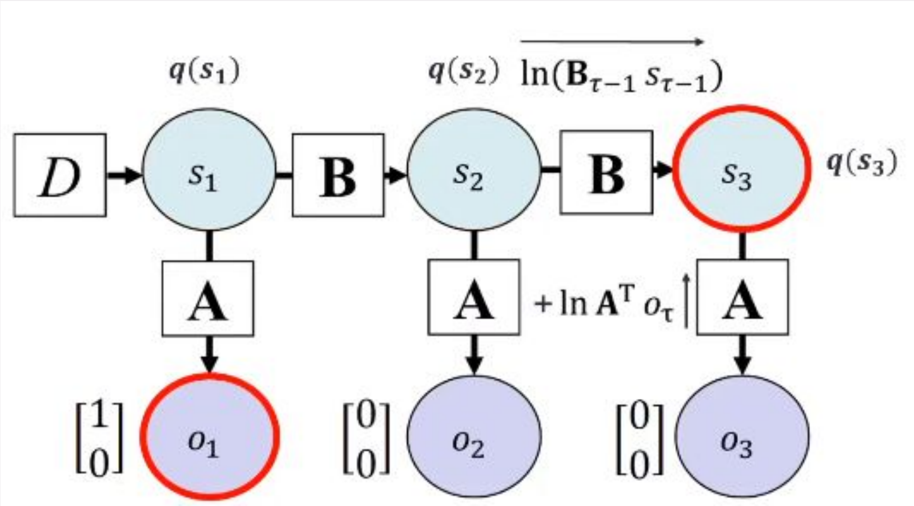



$$\mathbf{q}(s_1) = \sigma\left(\begin{bmatrix}-3.7\\-3.0\end{bmatrix}\right) = \begin{bmatrix}.55\\.45\end{bmatrix}$$

# How this shape work and what is F?



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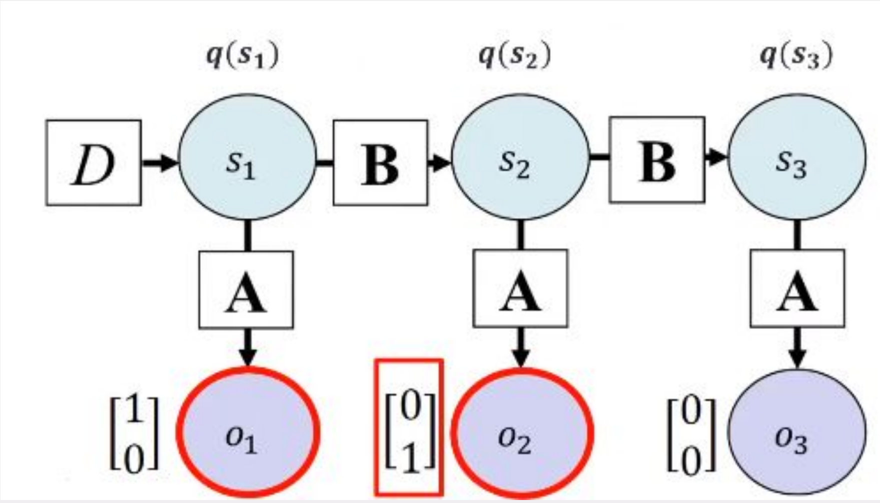


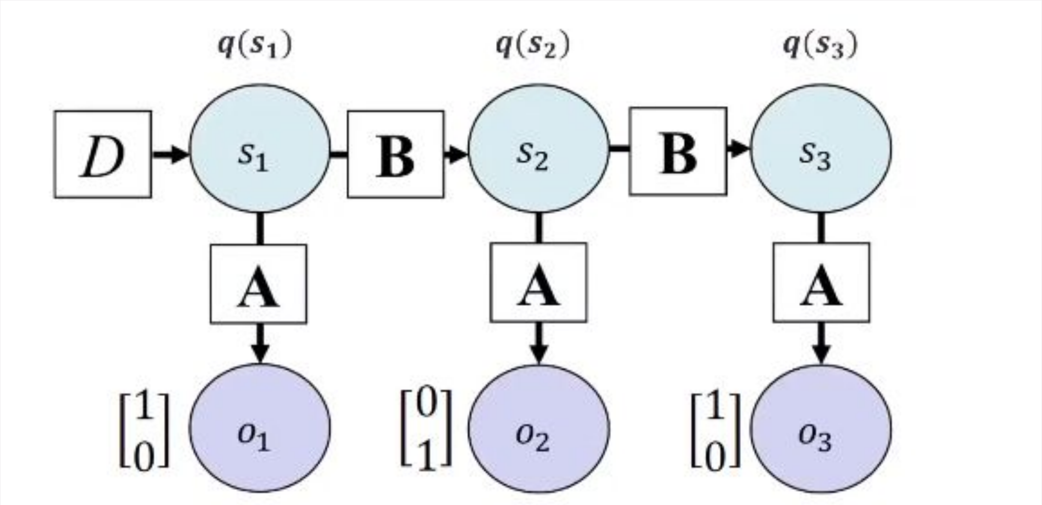


Static and dynamic perception

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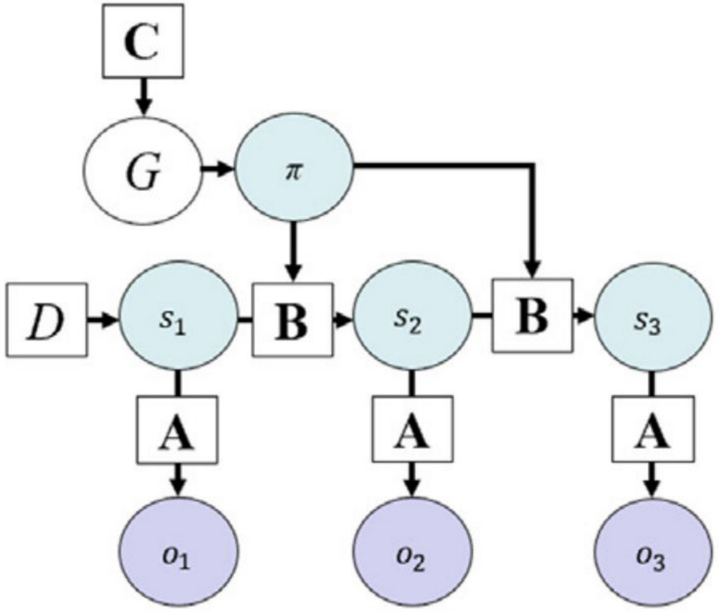
# Dynamic perception with policy selection

Each policy (π) entails a different sequence of actions, which corresponds to different transitions between states (i.e., different Bπ ,τ matrices). Based on expected free energy (G; which in turn depends on prior preferences, C), the highest probability will be assigned to policies expected to minimize uncertainty over states and maximize the probability of preferred observations.




Equations

$$s_{\pi,\tau=1} = \sigma\left(\frac{1}{2}\left(\ln D + \ln(\mathbf{B}^{\dagger}_{\pi,\tau}\, s_{\pi,\tau+1})\right) + \ln \mathbf{A}^{\mathrm{T}} o_\tau\right)$$

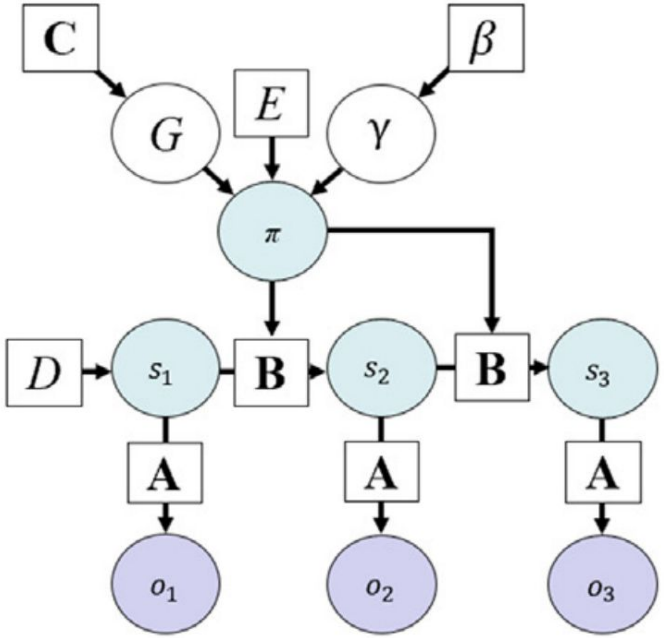
$$s_{\pi,\tau>1} = \sigma\left(\frac{1}{2}\left(\ln(\mathbf{B}_{\pi,\tau-1}\, s_{\pi,\tau-1}) + \ln(\mathbf{B}^{\dagger}_{\pi,\tau}\, s_{\pi,\tau+1})\right) + \ln \mathbf{A}^{\mathrm{T}} o_\tau\right)$$

$$G_\pi = \sum_\tau \left(\mathbf{A} s_{\pi,\tau} \cdot (\ln \mathbf{A} s_{\pi,\tau} - \ln \mathbf{C}_\tau) - diag(\mathbf{A}^{\mathrm{T}} \ln \mathbf{A}) \cdot s_{\pi,\tau}\right)$$

$$\pi = \sigma(-G)$$

## Dynamic perception with flexible policy selection

Introduces an expected free energy precision term $\gamma = E[\gamma]$
$\gamma$ represents the agent's confidence in policy selection




Additional equations

$$F_\pi = \sum_\tau s_{\pi,\tau} \cdot \left( \ln s_{\pi,\tau} - \frac{1}{2}\left( \ln(\mathbf{B}_{\pi,\tau-1}\, s_{\pi,\tau-1}) + \ln(\mathbf{B}^{\dagger}_{\pi,\tau}\, s_{\pi,\tau+1}) \right) - \ln \mathbf{A}^{\mathrm{T}} o_\tau \right)$$

$$\pi_0 = \sigma(\ln E - \gamma G)$$

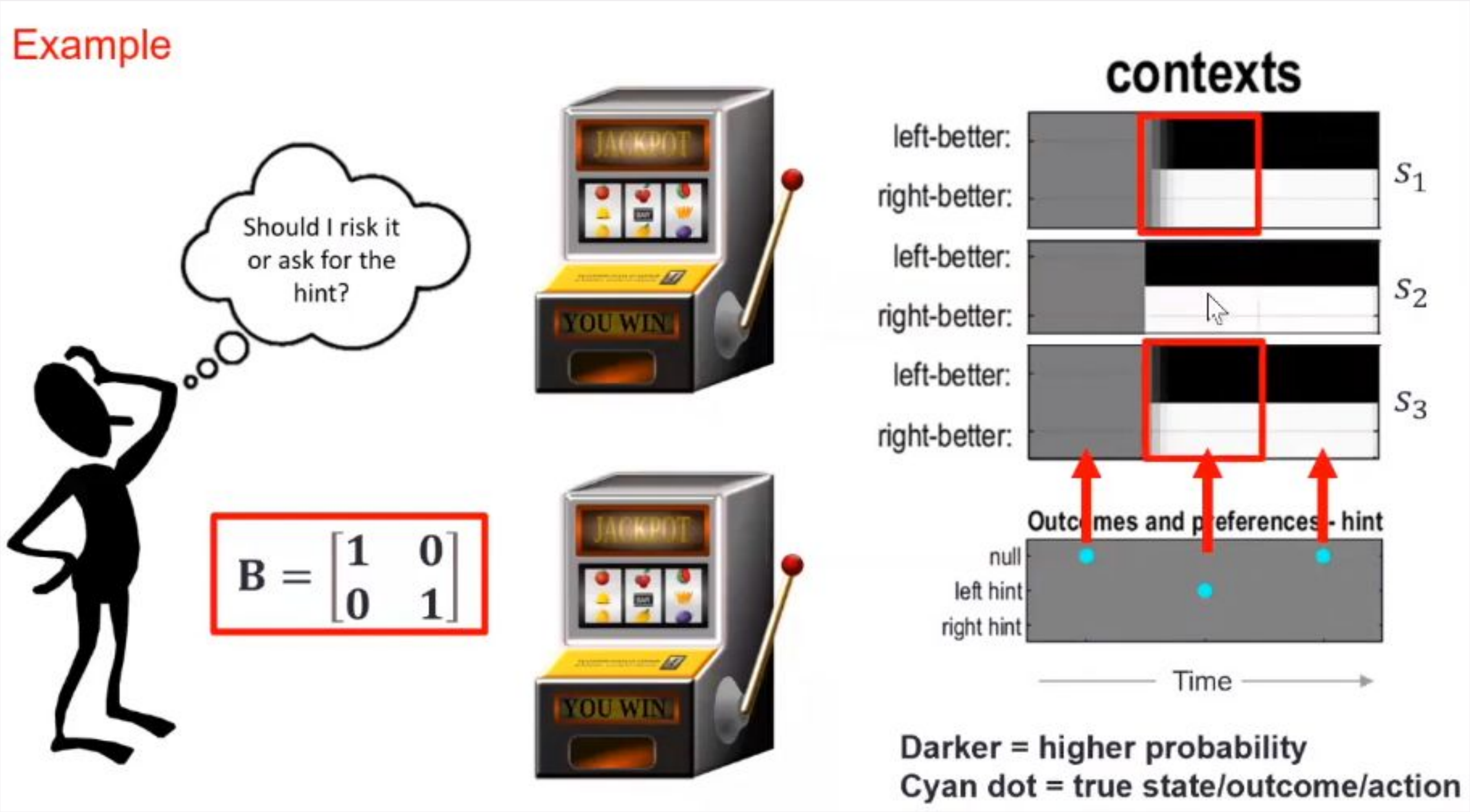
$$\pi = \sigma(\ln E - F - \gamma G)$$

$$p(\gamma) = \Gamma(1, \beta)$$

$$\mathrm{E}[\gamma] = \gamma = 1/\beta$$

$$\beta = \beta - \beta_{update}/\psi$$

$$\beta_{update} = \beta - \beta_0 + (\pi - \pi_0) \cdot (-G)$$


Example
Should I risk it or ask for the hint?
B = [1 0; 0 1]
JACKPOT
YOU WIN
JACKPOT
YOU WIN
contexts
left-better:
right-better:
left-better:
right-better:
left-better:
right-better:
s1
s2
s3
Outcomes and preferences - hint
null
left hint
right hint
Time
Darker = higher probability
Cyan dot = true state/outcome/action

# Active inference: Neural process theory

$$s_{\tau>1} = \sigma\left(\frac{1}{2}\left(\ln(\mathbf{B}_{\tau-1}\, s_{\tau-1}) + \ln(\mathbf{B}_{\tau}^{\dagger}\, s_{\tau+1})\right) + \ln \mathbf{A}^{\mathrm{T}}\, o_{\tau}\right)$$

Approximate posterior over states

$$\varepsilon_{\pi,\tau} \leftarrow \overbrace{\frac{1}{2}\left(\ln(\mathbf{B}_{\pi,\tau-1} s_{\pi,\tau-1}) + \ln(\mathbf{B}_{\pi,\tau}^{\dagger} s_{\pi,\tau+1})\right) + \ln\mathbf{A}^{\mathrm{T}} o_{\tau}}^{\text{Generative model after observation}} - \ln s_{\pi,\tau}$$

Membrane voltage

$$v_{\pi,\tau} \leftarrow v_{\pi,\tau} + \varepsilon_{\pi,\tau} \qquad v_{\pi,\tau} = \ln s_{\pi,\tau}$$

$$s_{\pi,\tau} \leftarrow \sigma(v_{\pi,\tau})$$

Normalized Firing rate

# Variational Free Energy

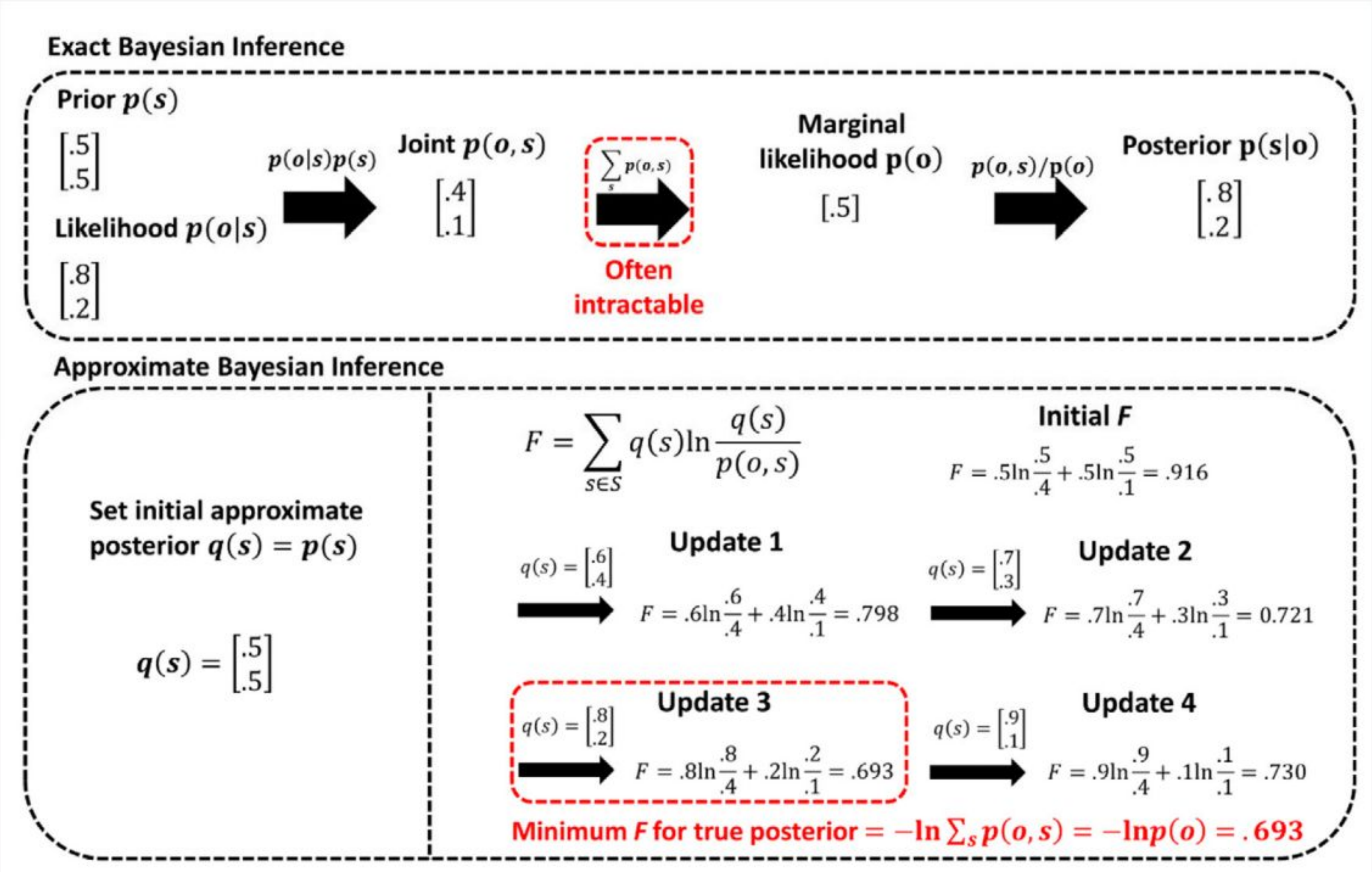
$$m = p(o, s) = p(o|s)p(s)$$

$$F = \boxed{\mathrm{E}_{q(s)}[\ln \frac{q(s)}{p(s|o)}]} - \boxed{\ln p(o)}$$

**Difference** between **approximate** and **true posterior**

"How **surprising** is this observation under my model?"

# Variational Free Energy

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# Variational Free Energy

**Complexity** + **Prediction error**

**Complexity** − **Accuracy**

$$F = \boxed{D_{KL}[q(s)||p(s)]} - \boxed{E_{q(s)}[\ln p(o|s)]}$$

"How much do beliefs need to change?"

"How accurate are my predictions?"

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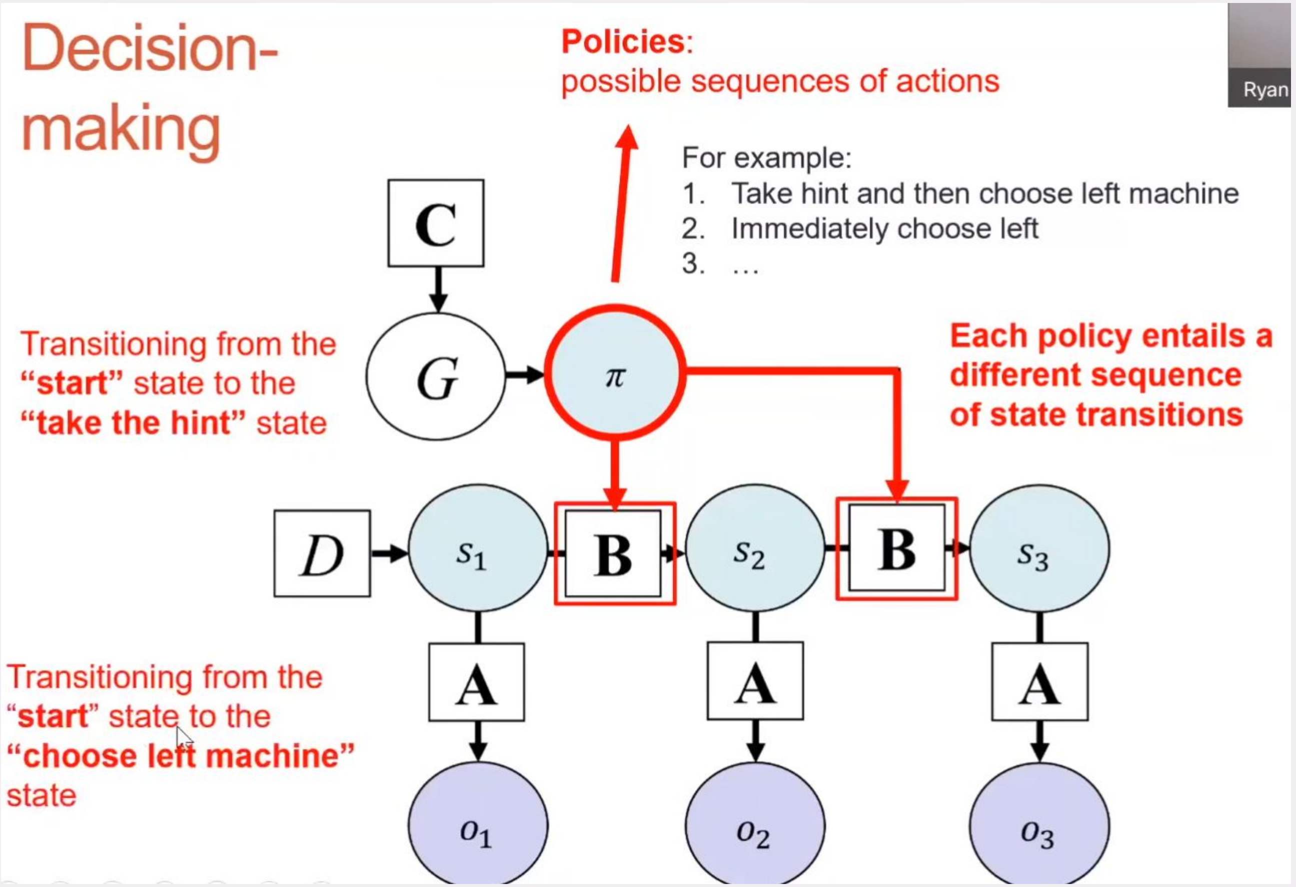
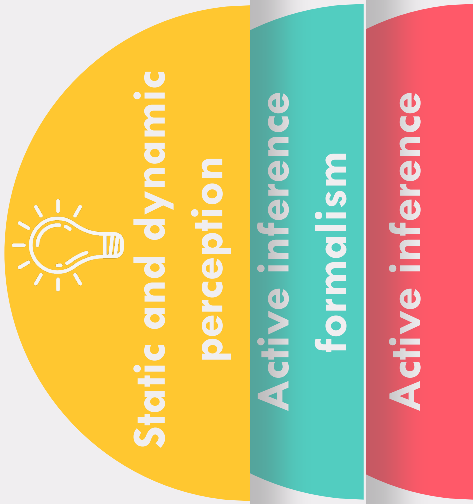
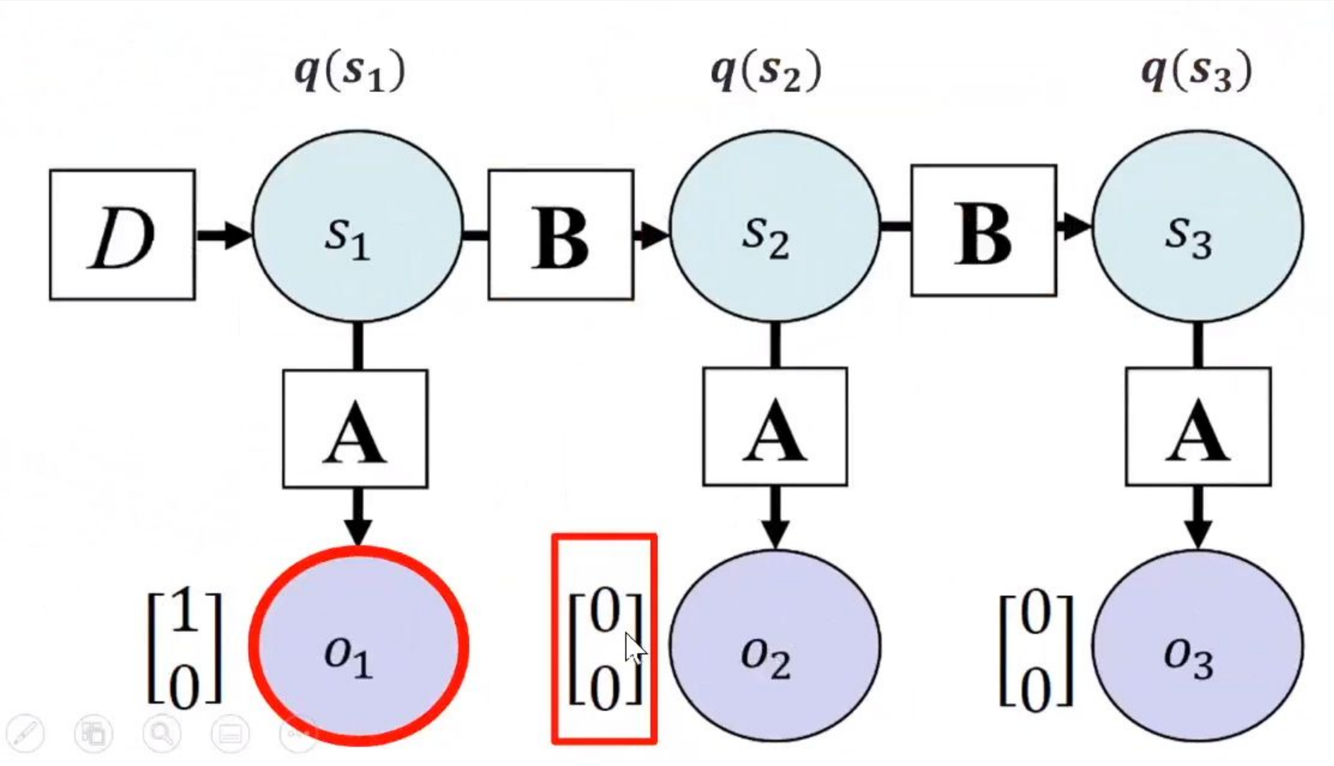
- Policies are state transitions that the agent has control over.(r or l T maze,..)
- Active inference agents by definition need to select policies that will minimize variation free energy.
- But, future variational free energy depends on observations that by definition have not yet happened.




$q(s_1)$
$q(s_2)$
$q(s_3)$
D
$s_1$
B
$s_2$
B
$s_3$
A
A
A
$\begin{bmatrix}1\\0\end{bmatrix}$
$o_1$
$\begin{bmatrix}0\\0\end{bmatrix}$
$o_2$
$\begin{bmatrix}0\\0\end{bmatrix}$
$o_3$

# Expected Free Energy (EFE)

The model assumes that the agent **optimizes** its beliefs by minimizing **Expected Free Energy (EFE)**, balancing two key factors:

### 1. Risk

The deviation from preferred outcomes. Agents navigate towards observations aligned with their preferences.

### 2. Ambiguity

Uncertainty about states. This **epistemic drive** motivates the agent to seek information gain.

**The C Vector: Encoding Preferences**

Represents a distribution that encapsulates favored states. For example, an agent seeking a comfortable body temperature might adopt a policy of staying indoors during cold winters.

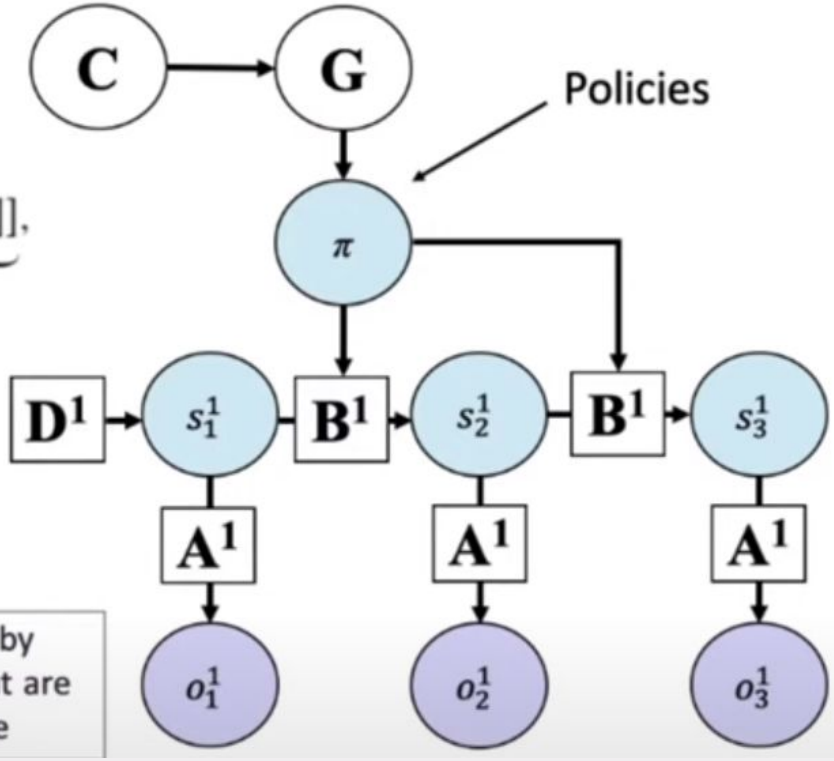
*"Policies with lowest EFE generate preferred outcomes while maximizing information gain."*

# Expected Free Energy

$$G(\pi,\tau) = \underbrace{D[Q(o_\tau|\pi)||P(o_\tau)]}_{\textit{expected cost}} + \underbrace{E_{\tilde{Q}}[H[P(o_\tau|s_\tau)]]}_{\textit{expected ambiguity}},$$

Minimize **deviation** between predicted and preferred outcomes **C**

Minimize ambiguity by moving to states that are the most informative

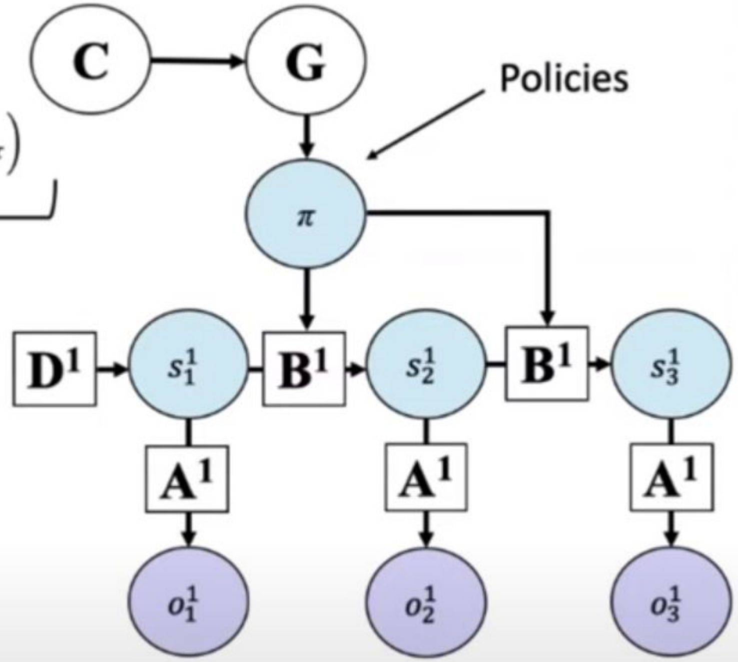


# Policy selection

$$\underbrace{F_\pi = \sum_\tau s_{\pi,\tau} \cdot \left(\ln s_{\pi,\tau} - \frac{1}{2}\left(\ln \mathbf{B}_{\pi,\tau-1}\, s_{\pi,\tau-1} + \ln \mathbf{B}^{\dagger}_{\pi,\tau} s_{\pi,\tau+1}\right) - \ln \mathbf{A}^T o_\tau\right)}_{\text{Variational free energy under each policy}}$$

$$\underbrace{G_\pi = \sum_\tau \left(o_{\pi,\tau} \cdot \left(\ln o_{\pi,\tau} - \ln \mathbf{C}\right) - diag(\mathbf{A}^T \ln \mathbf{A}) \cdot s_{\pi,\tau}\right)}_{\text{Expected free energy under each policy}}$$

$$\boxed{\boldsymbol{\pi} = \sigma(-F - G)}$$



# The implementation of learning in modeling multi-trial behavioral data

Learning implementation

Static and dynamic perception

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# Reference Materials

A step-by-step tutorial on **active inference** and its application to **empirical data**